NORTHERN GULF PORTS AUTHORITY · HEALTH, SAFETY & ENVIRONMENT DEPARTMENT

Port Equipment Safety & Environmental Standard

Document code	**PESE-03**
Version	2.0
Effective date	1 January 2026
Classification	Internal: may be shared with bidders

> **SAMPLE DOCUMENT.** Fictional standard created to test procurement-evaluation software.

1. Purpose and scope

1.1 This standard protects port workers, drivers, the public and the environment from hazards created by equipment installed in the port, during construction, operation, maintenance and decommissioning.

1.2 It applies to all fixed and relocatable equipment installed in NGPA terminals and to the works needed to install it, whether performed by NGPA staff or by suppliers and their subcontractors.

1.3 Requirements are identified by a code (e.g. SAF-01) and a priority: Mandatory requirements must be met; Important requirements are expected and weigh heavily in evaluations; Desirable requirements are good practice.

2. Definitions

Term	Meaning
Controlled area	An area where special radiation protection measures apply, delimited by physical barriers and signage.
ALARA	As Low As Reasonably Achievable: the principle of keeping exposure as low as reasonably achievable.
HSE officer	A qualified health, safety and environment professional with authority to stop unsafe work.

3. Radiation protection

These provisions apply to all equipment generating ionising radiation, including X-ray and linear-accelerator inspection systems.

3.1 **SAF-01** — No radiation-emitting equipment may be operated before it is licensed by the national radiation regulator. (Mandatory)

3.2 **SAF-02** — The annual dose at the boundary of any controlled area shall not exceed 1 mSv (public limit); occupational exposure shall be kept as low as reasonably achievable (ALARA). (Mandatory)

3.3 **SAF-03** — Occupied vehicle cabs shall never be exposed to the primary high-energy beam; any cab scanning shall use a low-dose mode delivering 0.25 µSv or less per pass. (Mandatory)

SAMPLE — fictional standard created for software testing.

3.4 **SAF-04** — Controlled areas shall be marked with bilingual (Arabic and English) radiation warning signs, beam-on warning lights and audible alarms. (Mandatory)

3.5 **SAF-05** — Emergency stop buttons, interlocked with the radiation source, shall be reachable within 10 metres from any point inside the controlled area. (Mandatory)

3.6 **SAF-06** — The supplier shall provide personal dosimeters for operators during the first 12 months of operation and train all operators in radiation protection. (Important)

4. Environmental resilience of equipment

Port terminals are exposed to high temperatures, humidity, salt spray and dust.

4.1 **SAF-07** — Outdoor equipment shall be rated for continuous operation from 0 °C to +50 °C and up to 95% relative humidity in a saline atmosphere, with electronics enclosures of at least IP54. (Mandatory)

4.2 **SAF-08** — Structural steel shall be protected for corrosivity category C5-M (ISO 12944) with a design durability of at least 15 years. (Important)

5. Noise and environmental protection

5.1 **SAF-09** — Noise from equipment, including HVAC units and generators, shall not exceed 70 dB(A) at 1 metre from operator cabins. (Important)

5.2 **SAF-10** — Hazardous waste (oils, batteries, lead shielding off-cuts and decommissioned sources) shall be removed by the supplier through licensed contractors, with disposal certificates. (Important)

6. Construction safety

6.1 **SAF-11** — Before civil works start, method statements and risk assessments shall be approved by the HSE Department, and the supplier shall keep a full-time HSE officer on site. (Important)

6.2 Hot work, work at height and excavation require a permit to work issued by the terminal operator.

7. Fire safety

7.1 **SAF-12** — Operator cabins and electrical rooms shall have fire detection; server and electrical rooms shall have clean-agent fire suppression. (Desirable)

8. Compliance verification and exceptions

8.1 Radiation surveys are performed at Site Acceptance Test and annually by a qualified expert.

8.2 The HSE Department may stop any work that does not comply with this standard.

Annex: requirements register

Code	Requirement	Priority

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SAF-01	No radiation-emitting equipment may be operated before it is licensed by the national radiation regulator.	Mandatory
SAF-02	The annual dose at the boundary of any controlled area shall not exceed 1 mSv (public limit); occupational exposure shall be kept as low as reasonably achievable (ALARA).	Mandatory
SAF-03	Occupied vehicle cabs shall never be exposed to the primary high-energy beam; any cab scanning shall use a low-dose mode delivering 0.25 µSv or less per pass.	Mandatory
SAF-04	Controlled areas shall be marked with bilingual (Arabic and English) radiation warning signs, beam-on warning lights and audible alarms.	Mandatory
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